SALES FORECASTING & DEMAND PREDICTION ANALYSIS

Role: Data Analyst Project

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1. Introduction

In today's competitive retail environment, forecasting sales and understanding customer demand is essential for effective business planning. This project focuses on analysing historical sales data to identify trends, seasonal patterns, and predict future demand.

The goal is to improve:

- Inventory management
- Sales planning
- Data-driven decision-making

2. Dataset Description

A dataset of **12,000+ records** was generated, containing:

- Date
- Product ID
- Product Category
- Region
- Units Sold

- Price
- Revenue

This dataset simulates real-world retail transactions across multiple regions and product categories.

3. Data Cleaning & Preprocessing

The dataset was cleaned and prepared using Python:

- Converted Date column to proper datetime format
- Removed duplicate records
- Checked for missing values
- Created additional columns:
  - Year
  - Month
  - Month Name

This ensured accurate and structured data for analysis.

4. Data Analysis

🔹 Sales Trend

- Sales show a **steady increasing trend over time**
- Minor fluctuations observed across months

🔹 Seasonal Patterns

- Clear **monthly variations** in sales
- Indicates strong **seasonal demand behaviour**

🔹 Product Category Performance

- Certain categories generate **higher revenue**
- Some categories underperform and need improvement

🔹 Regional Analysis

- Sales are **evenly distributed across regions**
- No single region dominates

5. Sales Forecasting

A forecasting model was built using **Linear Regression**.

📈 **Key Findings:**

- Sales expected to **increase slightly in the future**
- Demand remains **stable and consistent**
- No major decline predicted

6. Dashboard Overview

An interactive dashboard was developed using **Power BI**, including:

📊 **KPI Metrics:**

- Total Sales
- Total Units Sold
- Average Revenue

📈 **Visualizations:**

- Monthly Sales Trend
- Sales by Category
- Sales by Region
- Peak Sales Months
- Forecasted Sales Trend

🎛️ **Filters:**

- Year
- Region
- Product Category

7. Key Insights

- Total Sales ≈ **30.43M**
- Total Units Sold ≈ **119K**
- Sales trend is **increasing over time**
- Demand shows **clear seasonal patterns**
- Forecast indicates **stable future growth**

8. Business Recommendations

- Increase inventory before peak sales months
- Focus on high-performing product categories
- Improve marketing strategies for low-performing products
- Use forecasting for demand planning
- Align business strategies with seasonal trends

9. Conclusion

The analysis shows that the business is experiencing **steady growth with predictable demand patterns**. By leveraging forecasting and data insights, the company can significantly improve operational efficiency and maximize revenue.

Sales Forecasting & Demand Analysis Dashboard

119K
Total Units Sold

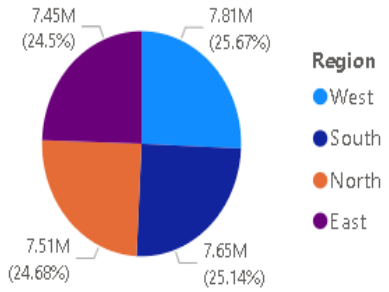
30.43M
Total Sales

2.54K
Average of Revenue

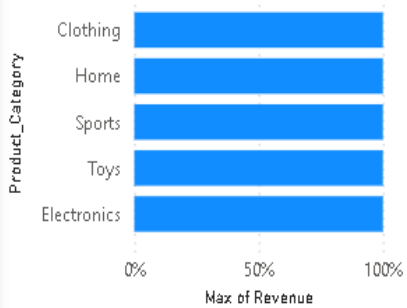
Monthly Sales Trend

Sales Forecast vs Actual

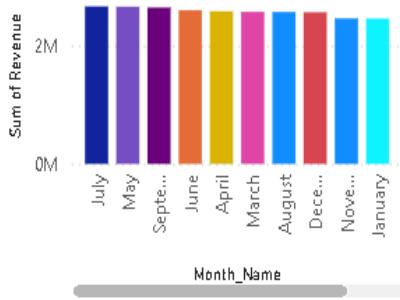
Revenue by Region



Top Performing Categories



Peak Sales Month



FILTERS

Region

East | South
North | West

ProductCategory

Clothing | Sports
Electronics | Toys
Home

Year

2022 | 2023 | 2024